

B1528 SIROCCO current-heading/rudder force comparison — 30 deg rudder limit

This interactive report visualizes **rudder-induced**, **OCIMF-inspired hull-current review**, and summed **total** force/yaw-moment components for B1528 SIROCCO current-heading/rudder combinations with rudder angles limited to -30 deg through +30 deg. Totals are comparison loads only; this is not a validated whole-vessel current-load, mooring, tug, bank-effect, or compliance model.

SIROCCO current-heading/rudder force-component sweep at COG; rudder-induced ship-fixed X/Y/N components are derived by rotating local current-frame loads by heading offset, first-cut OCIMF-inspired hull current review loads are estimated separately using transparent placeholder heading functions, and the report sums current+rudder resultants; bank effect, tug loads, mooring-line stiffness, current-profile variation, propeller race, and class compliance conclusion are excluded

Default speed policy: 4.56 kn is an extra chart-default case outside the requested engineering sweep list. Rows are flagged with `is_chart_default_extra_speed`.

Input data

Key input values used by both the interactive HTML and print/PDF report views.

LBP	225.500 m
Beam	32.260 m
Draft	12.200 m
Current frontal projected area	393.572 m²
Current lateral projected area	2751.100 m²
Yaw lever	135.300 m
Rudder area	44.939563 m²
Rudder span	9.000 m
Ship speed over ground	0.000 kn
Water density	1025.0 kg/m³
Barrass/workbook β	600.0
Propeller rotation factor Cr	1.0
Current speed sweep	1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5 kn + chart-default 4.56 kn
Heading/rudder grid	-10..10 deg heading × -30..30 deg rudder, 1 deg step

Sample calculation

Representative hand-check point: **chart-default 4.56 kn, heading 0 deg, rudder +1 deg, Cr=1.0.**

1. $V = 4.56 \text{ kn} \times 0.51444 = 2.34585 \text{ m/s}$; displayed as **4.56 kn = 2.34585 m/s.**

2. $\alpha = \delta - \psi = 1.0 \text{ deg} - 0.0 \text{ deg} = 1.0 \text{ deg}.$

3. $F = 600.0 \times 44.939563 \times 2.34585^2 \times 1.0 = 148381.324 \text{ N}.$

4. $F_n = F \times \sin(\alpha) = 2589.611 \text{ N}.$

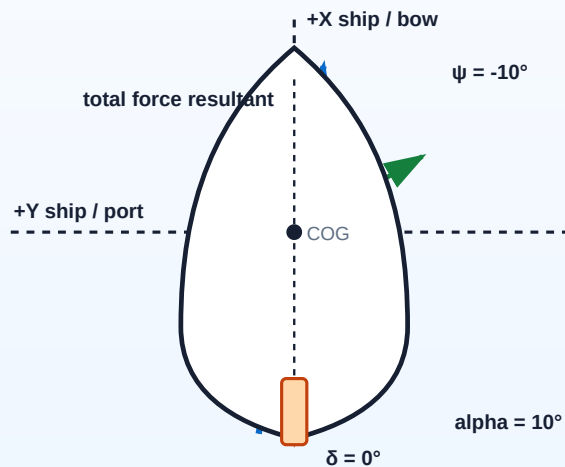
5. $X_{\text{ship}} = 45.195 \text{ N}, Y_{\text{ship}} = 2589.217 \text{ N}, N_{\text{ship}} = 350.321028 \text{ kN-m}.$

6. $q = 0.5 \times \rho \times V^2$; current coefficients use $C_x=1.05 \times \text{abs}(\cos(\psi))$, $C_y=\sin(\psi)$, $C_m=0.55 \times \sin(\psi)$ as transparent placeholder heading functions.

7. $X_{\text{current}} = 1165484.513 \text{ N}, Y_{\text{current}} = 0.000 \text{ N}, N_{\text{current}} = 0.000000 \text{ kN-m}$; totals are component sums at COG.

Ship/current/rudder geometry schematic

This schematic updates with the current speed/rudder controls and uses the selected breakdown heading. It shows ship-fixed axes, the current heading offset ψ , rudder command δ , and effective rudder inflow angle $\alpha = \delta - \psi$. Positive bow-to-port convention applies to ψ , δ , Y_{ship} , and yaw moment.



- **Black dashed axes:** ship-fixed COG frame; +X forward and +Y port.
- **Blue vector:** current heading offset ψ relative to ship centerline.
- **Orange vector:** selected rudder command δ ; the selector intentionally controls this angle for Chart 1/3/4 review.
- **Green vector:** selected-case total horizontal force direction from current + rudder components.
- **Effective inflow:** $\alpha = \delta - \psi$; when alpha is zero, rudder-induced load is zero but hull-current load remains.

Speed

4.56 kn

Heading ψ

-10 deg

Rudder δ

0 deg

Effective alpha

10 deg

Controls

Current speed

Rudder angle for Chart 1

4.56 kn (chart default extra) ▼

0.0 deg ▼

Selected-speed envelope summary

Envelope values below update with the current-speed selector and use the full heading × rudder grid at that speed. This panel is intended to keep the chart default **4.56 kn** visible in screenshots while preserving requested sweep row counts.

Metric	Envelope case	Value
Selected speed	chart-default extra plane	4.56 kn · 1281 rows
Max Y_ship at selected speed	heading -4 deg · rudder 30 deg	65.384 kN
Max N_ship at selected speed	heading -4 deg · rudder 30 deg	8846.487 kN-m
Max resultant horizontal force	heading -10 deg · rudder 30 deg	95.378 kN

Chart 1 — Rudder-induced ship-fixed force components over heading

Shows rudder-induced X_{ship} , Y_{ship} , and resultant horizontal force in kN for the selected current speed and rudder angle. All 21 rudder angles are selectable; the default rudder angle is neutral **0.0 deg**.

Rudder-induced ship-fixed force component (kN) · current 4.56 kn · rudder 0 deg

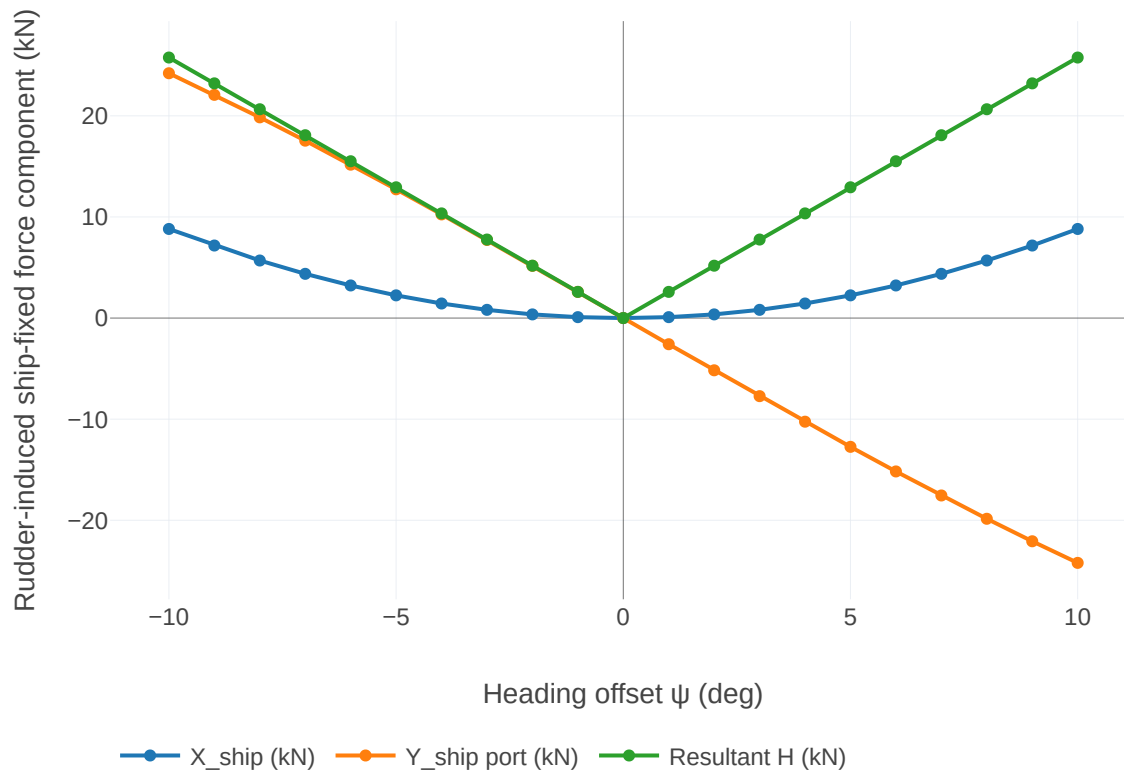


Chart 2 — Rudder-induced ship-fixed yaw moment heatmap over heading × rudder

Shows signed rudder-induced N_{ship} yaw moment (kN-m) over heading offset and rudder angle for the selected current speed.

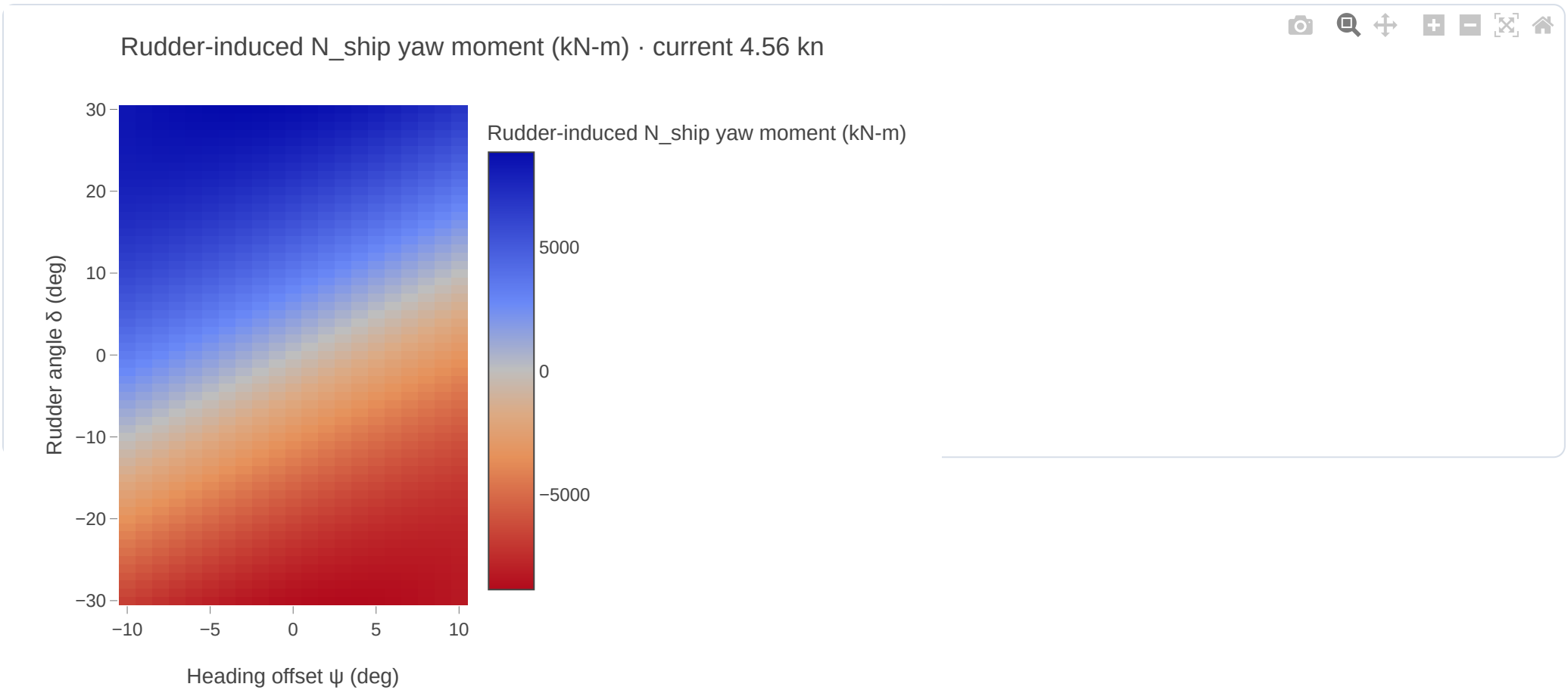


Chart 3 — OCIMF-inspired current vs rudder vs total horizontal resultant

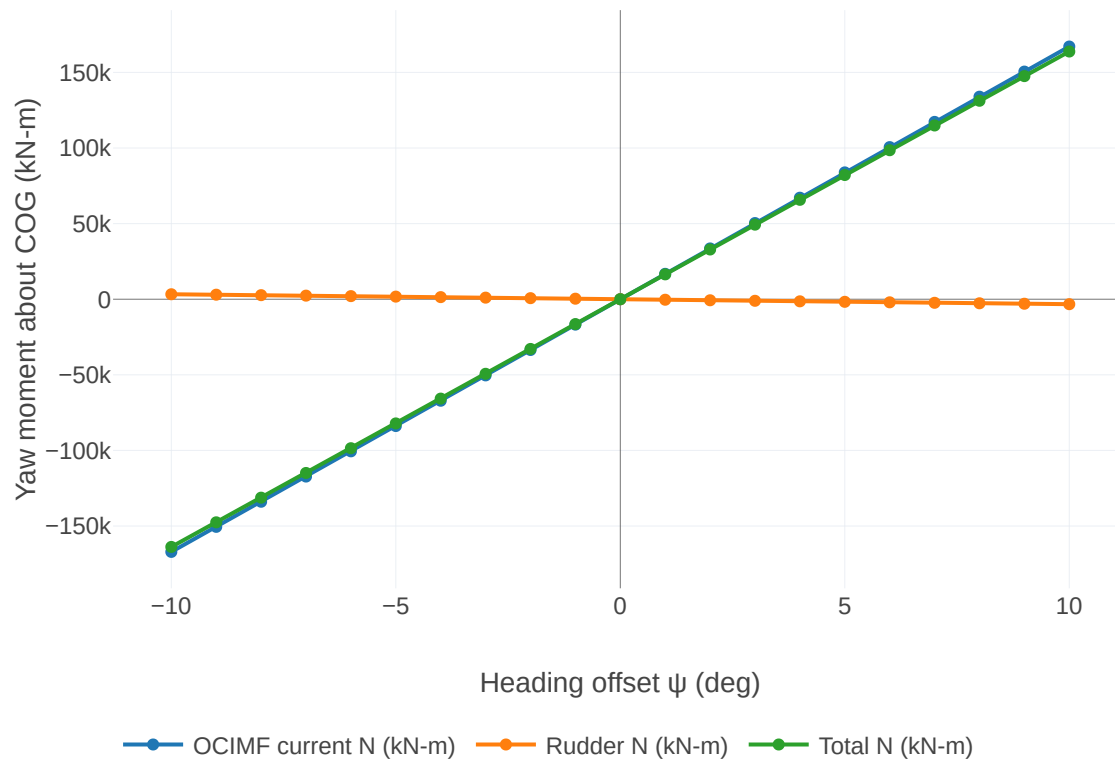
Compares individual horizontal resultants from the first-cut OCIMF-inspired hull-current review estimate and rudder-induced component, plus the vector-summed total resultant in ship-fixed COG axes for the selected speed and rudder angle. Current coefficients are transparent placeholder heading functions, not vessel-specific OCIMF curves.



Chart 4 — OCIMF-inspired current vs rudder vs total yaw moment about COG

Compares signed yaw moment about COG from the OCIMF-inspired current-review component, the rudder-induced component, and the summed total yaw moment. Positive yaw moment is bow-to-port.

current vs rudder vs total yaw moment about COG (kN-m) · current 4.56 kn · rudder



Selected-case force breakdown

Updates with current speed and rudder angle. Values are shown at the heading with maximum absolute total yaw moment for the selected trace, so individual and resultant force paths can be reviewed together.

Component	X ship (kN)	Y ship port (kN)	Horizontal resultant (kN)	Yaw moment about COG (kN-m)
OCIMF current	1147.778	-1347.316	1769.931	-167100.926
Rudder induced	8.813	24.212	25.766	3275.918
Total / mooring reaction opposite	1156.591	-1323.104	1757.358	-163825.008

Selected breakdown case: current 4.56 kn · heading -10 deg · rudder 0 deg · total yaw reaction for mooring context = 163825.008 kN-m.

Method and provenance

```
V=kn*0.51444; F=beta*A_R*V^2*Cr; alpha=delta-psi; X_local=F*sin(alpha)^2; Y_local=F*sin(alpha)*cos(alpha);  
X_ship=X_local*cos(psi)-Y_local*sin(psi); Y_ship=X_local*sin(psi)+Y_local*cos(psi); N=Y_ship*yaw_lever
```

- Heading/rudder effective-angle convention: $\alpha = \text{rudder_angle_deg} - \text{heading_offset_deg}$.
- Local-to-ship transform rotates local current-frame X/Y loads into ship-fixed COG axes.
- Current review equations: $q=0.5*\rho*V^2$; $C_x=1.05*\text{abs}(\cos(\psi))$; $C_y=\sin(\psi)$; $C_m=0.55*\sin(\psi)$; $X_{\text{current}}=q*(\text{beam}*draft)*C_x$; $Y_{\text{current}}=q*(LBP*draft)*C_y$; $N_{\text{current}}=q*(LBP*draft)*LBP*C_m$.
- When rudder_angle_deg equals $\text{heading_offset_deg}$, α is zero and this rudder-induced component is zero; that is not total hull current load.

- Scope note: hull current loads are first-cut OCIMF-inspired comparison loads using placeholder heading functions only; this is not a validated whole-vessel current-load or oblique-current hull/rudder interaction model.

Summary

```
{
  "row_count": 11529,
  "requested_engineering_row_count": 10248,
  "extra_default_row_count": 1281,
  "max_abs_ship_sway_force_N": 65384.23353202306,
  "max_abs_ship_sway_signed_force_N": 65384.23353202306,
  "max_abs_ship_sway_case": "current 4.56 kn, heading -4 deg, rudder 30 deg",
  "max_abs_yaw_moment_kN_m": 8846.48679688272,
  "max_abs_yaw_signed_moment_kN_m": 8846.48679688272,
  "max_abs_yaw_case": "current 4.56 kn, heading -4 deg, rudder 30 deg",
  "max_resultant_horizontal_force_N": 95377.67650976167,
  "max_resultant_case": "current 4.56 kn, heading -10 deg, rudder 30 deg",
  "max_ocimf_current_resultant_horizontal_force_N": 1769931.1993048876,
  "max_ocimf_current_resultant_case": "current 4.56 kn, heading -10 deg, rudder -30 deg",
  "max_total_resultant_horizontal_force_N": 1813872.9536408707,
  "max_total_resultant_case": "current 4.56 kn, heading -10 deg, rudder -30 deg",
  "max_abs_total_yaw_moment_kN_m": 173863.00417282406,
  "max_abs_total_yaw_signed_moment_kN_m": -173863.00417282406,
  "max_abs_total_yaw_case": "current 4.56 kn, heading -10 deg, rudder -30 deg"
}
```